Adder Comparison

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Viet Ha ECEN 603

Agenda

- Question
- Background Adders
- Testing Methodology/Methods
- Results
- Conclusions
- Improvements.

Question:

- What type of adder will be the best in terms of speed,power and area for 8-bit adding.
- Based on the hybrid adder paper to get the adder and resources for it.
- Adders used CSA,CSELA,CLA,BKA,LFA,KSA, and pipelined versions of BKA(Brent-Kung Adder),LFA(Ladner-Fisher),KSA(Kogge Stone).
- Different from the hybrid adder because testing using 8-bit numbers and including pipelined version instead of pure combinational or Parallel.

Background Adders Carry Select Adders:

- Is A Parallel Adder
- Pre Computes The sum by with Cin=1 or 0
- Then Selects it based on the Actual Cin using Mux

A. Carry Select Adder (CSELA)

A carry select adder (CSELA) is a parallel adder that can simultaneously add two -bit values. Carry select adder is thought to be more efficient than carry look ahead adder, and it also operates quickly.

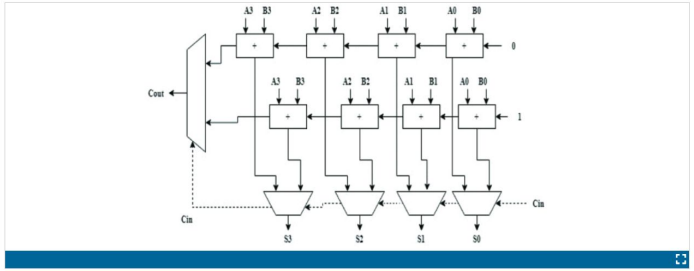


Fig. 1.
Carry select adder block adder.

Background Adders: Carry Save Adder

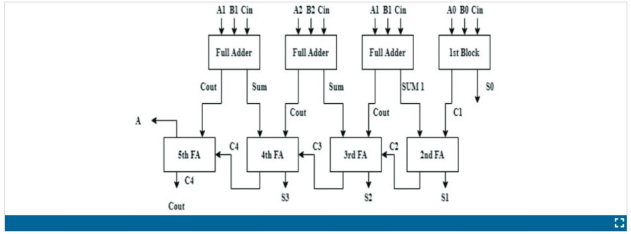
- Type of Parallel Adder
- Saves the Carry By Saving with other Adders
- Combined With CLA to make for Fast Adder

B. Carry Save Adder (CSA)

Another kind of parallel RCA is the Carry Save Adder, which does not compute the carry during the stages but instead stores or saves it and then calculates it later. Let us pretend we have to figure out A, B, and C. As a consequence, CSA separates this into the following equation:

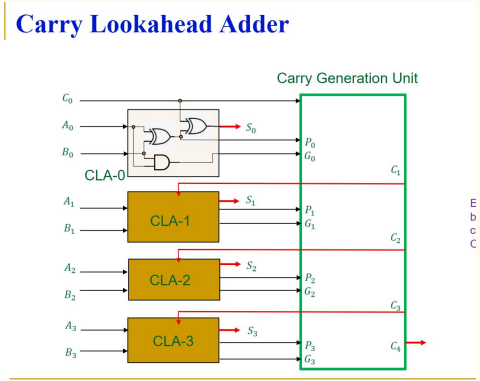
$$A + B + C = S + C \tag{2}$$

View Source



Background Adder Carry Lookahead Adder:

- Type of Parallel Adder that Precomputes Carry
- Using Carry Generation unit with Generate and Prop Sig
- Avoid the Ripple Effect for Ripple Carry Adder.



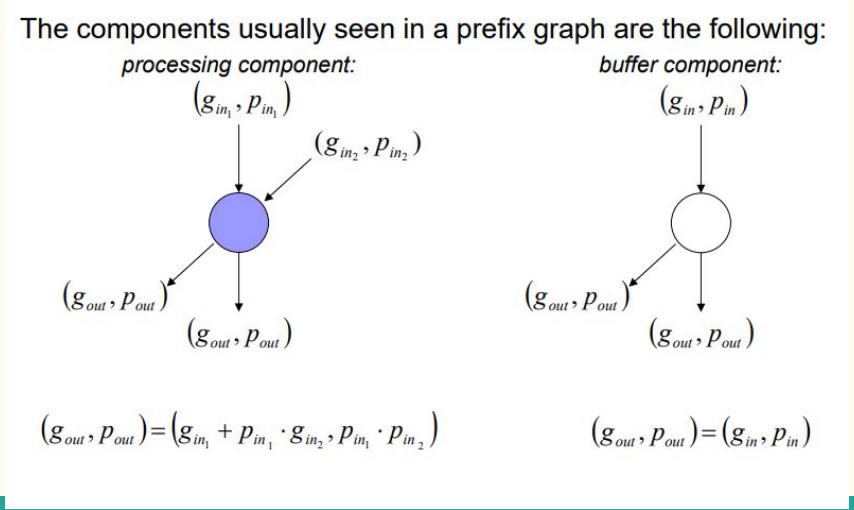
Background Parallel Prefix Adder:

- Special Class of Adder where it uses Graph to calculate Carry Sig
- Shown is how Calculate the Carry and Propagate Signals
- Original Carry In for the first ci.
- Different Adders have different Tree's

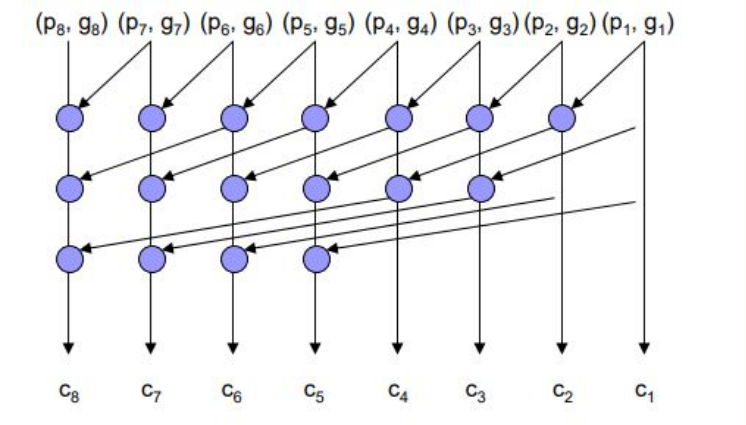
$$p_i = x_i \oplus y_i$$

$$g_i = x_i \bullet y_i$$

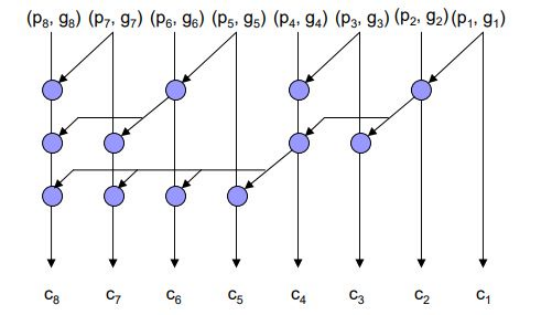
$$c_i = g_i + p_i \cdot c_{i-1} = x_i \cdot y_i + (x_i \oplus y_i) \cdot c_{i-1}$$



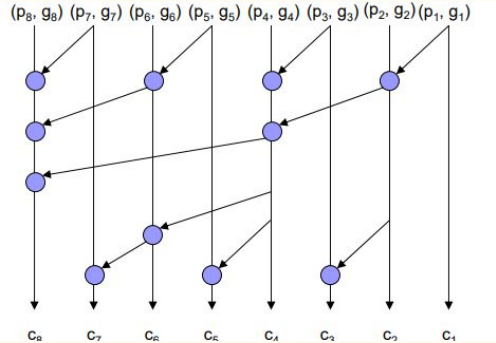
BKA,LFA,KSA Tree Structures.



KSA



LFA



BKA

Testing Methodology

- Made Testbench to Simulate All Additions with every possible Input and Compared to Output
- Use exact load and clock delay to make sure everything was equal
- Did medium Effort to Prevent compiler from Optimizing away.

- create_clock -name "my_clk" -period 10 [get_ports clk]
- set_clock_latency 0.3 clk
- set_output_delay 1.7 -clock clk [all_outputs]
- set_load 0.1 [all_outputs]
- set_max_fanout 1 [all_inputs]
- set_fanout_load 8 [all_output]
- compile -map_effort medium -area_effort medium

Methods Pipelined:

- For all Parallel Prefix adders
- Added Pipelined Register for Carry,Propagate,Generate
- Then Added Another register for Nodes in Trees
- ~ roughly Three Clock Cycles For calculating Sum
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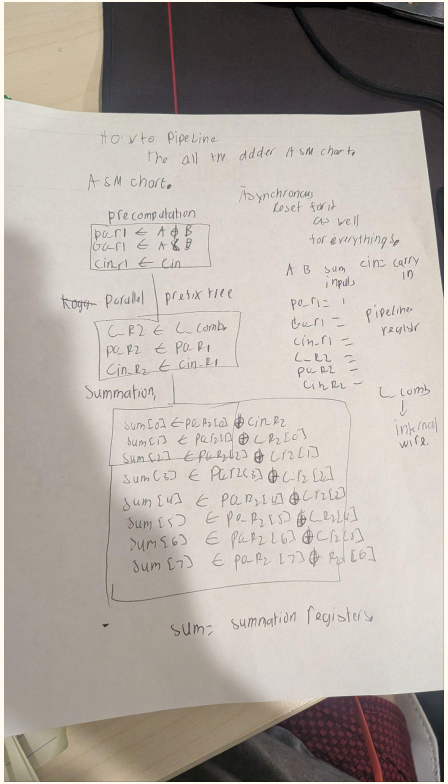
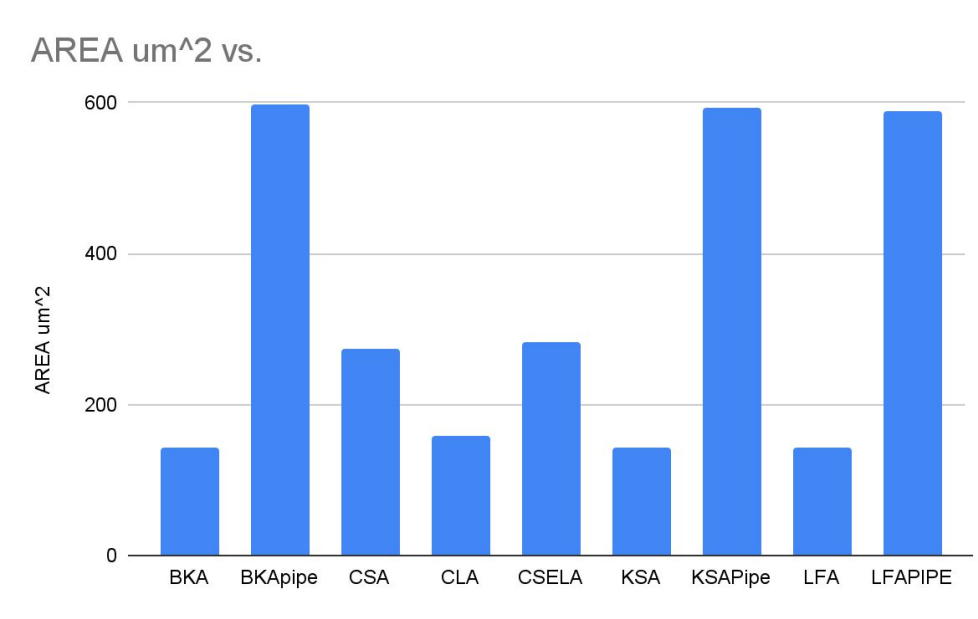


Results:


AREA um^2 vs.
AREA um^2
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400
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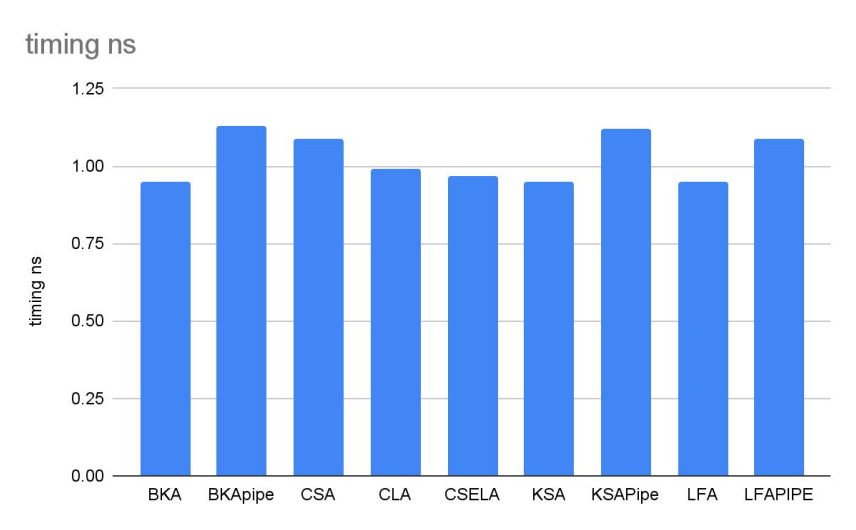
BKA
BKApipe
CSA
CLA
CSELA
KSA
KSAPipe
LFA
LFAPIPE


timing ns
timing ns
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1.00
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BKA
BKApipe
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CSELA
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KSAPipe
LFA
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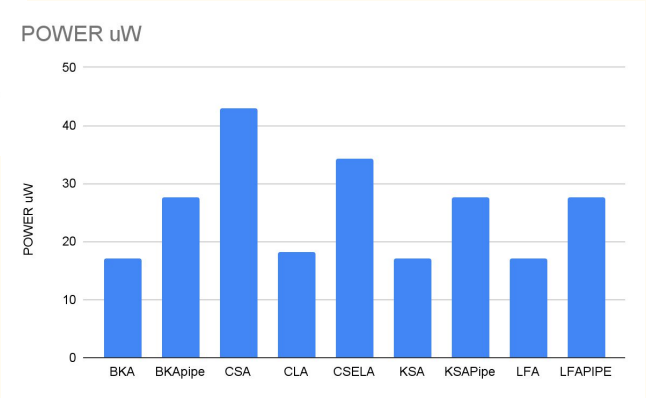
POWER uW
POWER uW
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30
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BKA
BKApipe
CSA
CLA
CSELA
KSA
KSAPipe
LFA
LFAPIPE

Conclusion:

- Overall Not worth it to pipeline the adders
- Nearly Doubles the area while adding more power and potentially slower.
- However the slack time could be beneficial as you are able to figure out how the slacks will work for it.
- Found out that Parallel Prefix Adders were the Fastest,however the Different Tree structures didn't matter
- Possibly due to Synopsys Compiler Optimizing Certain Functions away.

Improvements

- Possibly Include The Hybrid Adders for 16 bit to see which combination of adders work to increase performance.
- Why the BKA,CSA,LFA non pipelined is exact same when synthesized
- Optimize the Pipelining so that it actively decreases delay instead of increasing without size overhead.
- Implement the Han-Carlson Adder and potentially the pipelined version.

Citations:

R. R, B. Biju, A. Joy, V. Ganeshan and S. R, "Design and Performance Analysis of Various 32-bit Hybrid Adders using Verilog," 2022 IEEE International Conference on Signal Processing, Informatics, Communication and Energy Systems (SPICES), THIRUVANANTHAPURAM, India, 2022, pp. 105-112, doi: 10.1109/SPICES52834.2022.9774131. keywords: {Performance evaluation;Very large scale integration;Signal processing;SPICE;Hybrid power systems;Software;Delays;3-2 compressor;Brent-Kung adder (BKA);Han-Carlson adder (HCA);Hybrid adders;Kogge-Stone adder (KSA);Ladner-Fischer adder (LFA) and Parallel prefix adders},

[1] A. Al-Khalili, "Parallel prefix adders presentation," Lecture notes for COEN 6501: Digital Design and Synthesis, Dept. of Electrical and Computer Engineering, Concordia University, Montreal, QC, Canada. [Online]. Available: https://users.encs.concordia.ca/~asim/COEN_6501/Lecture_Notes/Parallel%20prefix%20adders%20presentation.pdf. Accessed: Mar. 21, 2026.